MAST30027: Modern Applied Statistics

Assignment 1, 2026 — Solutions

Table of contents

Question 1: Titanic survival from first principles
Question 2: Logistic regression with Antarctic penguins
Question 3: Gamma theory and New York ozone modelling

Question 1: Titanic survival from first principles

Data preparation

```
data(Titanic)

titanic_long <- as.data.frame(Titanic)
titanic_grouped <- reshape(
  titanic_long,
  idvar = c("Class", "Sex", "Age"),
  timevar = "Survived",
  direction = "wide"
)
titanic_grouped$Total <-
  titanic_grouped$Freq.No + titanic_grouped$Freq.Yes
titanic_grouped <- subset(
  titanic_grouped,
  Class == "3rd" & Total > 0
)
titanic_grouped$female <-
  as.integer(titanic_grouped$Sex == "Female")
titanic_grouped$adult <-
  as.integer(titanic_grouped$Age == "Adult")
titanic_grouped
```

```
   Class    Sex   Age Freq.No Freq.Yes Total female adult
3    3rd   Male Child      35       13    48      0     0
7    3rd Female Child      17       14    31      1     0
11   3rd   Male Adult     387       75   462      0     1
15   3rd Female Adult      89       76   165      1     1
```

(a) Maximum likelihood estimation

Let $\eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}$, where $\mathbf{x}_i = (1, \texttt{female}_i, \texttt{adult}_i)^\top$, and let $p_i = \Phi(\eta_i)$. Apart from terms that do not depend on $\boldsymbol{\beta}$, the grouped-binomial log-likelihood is

$$\ell(\boldsymbol{\beta}) = \sum_i [y_i \log\{\Phi(\eta_i)\} + (m_i - y_i) \log\{1 - \Phi(\eta_i)\}].$$

The following code maximises this log-likelihood directly. Small probability values are bounded away from zero and one to prevent numerical evaluation of $\log(0)$.

```
y <- titanic_grouped$Freq.Yes
m <- titanic_grouped$Total
X <- cbind(intercept = 1,
           female = titanic_grouped$female,
           adult = titanic_grouped$adult)

log_likelihood <- function(beta, X, y, m) {
  eta <- X %*% beta
  p <- pnorm(eta)
  p <- pmin(pmax(p, .Machine$double.eps),
            1 - .Machine$double.eps)

  return(sum(y * log(p) + (m - y) * log(1 - p)))
}

probit_fit <- optim(
  par = rep(0, ncol(X)),
  fn = log_likelihood,
  X = X,
  y = y,
  m = m,
  control = list(fnscale = -1),
  hessian = TRUE
)

beta_hat <- probit_fit$par
names(beta_hat) <- colnames(X)
beta_hat
```

```
 intercept     female      adult
-0.7579554  0.8307566 -0.2083906
```

```
probit_fit$convergence
```

```
[1] 0
```

```
eigen(probit_fit$hessian, symmetric = TRUE)$values
```

```
[1]  -20.72662  -79.05515 -692.54008
```

The estimates are approximately $\widehat{\beta}_0 = -0.758$, $\widehat{\beta}_1 = 0.831$, and $\widehat{\beta}_2 = -0.208$. The optimiser returned convergence code 0. All eigenvalues of the Hessian are negative, supporting that the stationary point is a local maximum.

(b) Fisher information and confidence intervals

Suppose

$$Y_i \sim \text{Binomial}(m_i, p_i), \qquad p_i = \Phi(\eta_i), \qquad \eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}.$$

For observation i, ignoring terms that do not depend on $\boldsymbol{\beta}$,

$$\ell_i = y_i \log p_i + (m_i - y_i) \log(1 - p_i).$$

Since

$$\frac{\partial p_i}{\partial \boldsymbol{\beta}} = \phi(\eta_i)\mathbf{x}_i,$$

the first derivative is

$$\begin{aligned} \frac{\partial \ell_i}{\partial \boldsymbol{\beta}} &= \left[\frac{y_i}{p_i} - \frac{m_i - y_i}{1 - p_i}\right] \frac{\partial p_i}{\partial \boldsymbol{\beta}} \\ &= \frac{y_i - m_i p_i}{p_i(1 - p_i)} \phi(\eta_i)\mathbf{x}_i. \end{aligned}$$

Define

$$A_i = \frac{\phi(\eta_i)}{p_i(1 - p_i)}.$$

Then

$$\frac{\partial \ell_i}{\partial \boldsymbol{\beta}} = (y_i - m_i p_i) A_i \mathbf{x}_i.$$

Differentiating again with respect to $\boldsymbol{\beta}$ gives

$$\frac{\partial^2 \ell_i}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}^\top} = \mathbf{x}_i \frac{\partial}{\partial \boldsymbol{\beta}^\top} [(y_i - m_i p_i) A_i].$$

Using the product rule,

$$\frac{\partial^2 \ell_i}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}^\top} = \mathbf{x}_i \left[\frac{\partial (y_i - m_i p_i)}{\partial \boldsymbol{\beta}^\top} A_i + (y_i - m_i p_i) \frac{\partial A_i}{\partial \boldsymbol{\beta}^\top}\right].$$

Now,

$$\frac{\partial (y_i - m_i p_i)}{\partial \boldsymbol{\beta}^\top} = -m_i \frac{\partial p_i}{\partial \boldsymbol{\beta}^\top} = -m_i \phi(\eta_i) \mathbf{x}_i^\top.$$

Therefore,

$$\begin{aligned} \frac{\partial^2 \ell_i}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}^\top} &= -m_i \phi(\eta_i) A_i \mathbf{x}_i \mathbf{x}_i^\top \\ &\quad + (y_i - m_i p_i) \mathbf{x}_i \frac{\partial A_i}{\partial \boldsymbol{\beta}^\top}. \end{aligned}$$

Substituting

$$A_i = \frac{\phi(\eta_i)}{p_i(1 - p_i)},$$

we obtain

$$\begin{aligned} \frac{\partial^2 \ell_i}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}^\top} &= -m_i \frac{\phi(\eta_i)^2}{p_i(1 - p_i)} \mathbf{x}_i \mathbf{x}_i^\top \\ &\quad + (y_i - m_i p_i) \mathbf{x}_i \frac{\partial}{\partial \boldsymbol{\beta}^\top} \left[\frac{\phi(\eta_i)}{p_i(1 - p_i)}\right]. \end{aligned}$$

Taking expectations,

$$\mathrm{E}(Y_i - m_i p_i) = 0,$$

so the second term vanishes. Hence,

$$-\mathrm{E}\left[\frac{\partial^2 \ell_i}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}^\top}\right] = m_i \frac{\phi(\eta_i)^2}{p_i(1 - p_i)} \mathbf{x}_i \mathbf{x}_i^\top.$$

Since

$$p_i = \Phi(\eta_i),$$

the Fisher information contribution from observation i is

$$\mathcal{I}_i(\boldsymbol{\beta}) = m_i \frac{\phi(\eta_i)^2}{\Phi(\eta_i)[1 - \Phi(\eta_i)]} \mathbf{x}_i \mathbf{x}_i^\top.$$

Summing over all independent observations,

$$\mathcal{I}(\boldsymbol{\beta}) = \sum_{i=1}^n m_i \frac{\phi(\eta_i)^2}{\Phi(\eta_i)[1 - \Phi(\eta_i)]} \mathbf{x}_i \mathbf{x}_i^\top.$$

Equivalently,

$$\mathcal{I}(\boldsymbol{\beta}) = \mathbf{X}^\top \mathbf{W} \mathbf{X},$$

where

$$W_{ii} = m_i \frac{\phi(\eta_i)^2}{\Phi(\eta_i)[1 - \Phi(\eta_i)]}.$$

```
eta_hat <- drop(X %*% beta_hat) # drop deletes the dimension of an array with length 1
p_hat <- pnorm(eta_hat)
weights_hat <- m * dnorm(eta_hat)^2 / (p_hat * (1 - p_hat))

fisher_information <- t(X) %*% diag(weights_hat) %*% X
variance_beta <- solve(fisher_information)
standard_errors <- sqrt(diag(variance_beta))
z975 <- qnorm(0.975)

coefficient_results <- data.frame(
  estimate = beta_hat,
  standard_error = standard_errors,
  lower_95 = beta_hat - z975 * standard_errors,
  upper_95 = beta_hat + z975 * standard_errors
)
round(coefficient_results, 4)
```

```
          estimate standard_error lower_95 upper_95
intercept  -0.7580         0.1580  -1.0676  -0.4483
female      0.8308         0.1115   0.6122   1.0493
adult      -0.2084         0.1608  -0.5236   0.1068
```

The female coefficient is positive, while its confidence interval excludes zero. The adult interval contains zero after sex has been included in this additive model.

(c) Likelihood-ratio test

We test $H_0 : \beta_1 = 0$ against $H_1 : \beta_1 \neq 0$, with the adult term in both models. Under H_0, the likelihood-ratio statistic is asymptotically χ_1^2.

```
X_reduced <- cbind(
  intercept = 1,
  adult = titanic_grouped$adult
)

reduced_fit <- optim(
  par = rep(0, ncol(X_reduced)),
  fn = log_likelihood,
  X = X_reduced,
  y = y,
  m = m,
  control = list(fnscale = -1)
)

LR_statistic <- 2 * (probit_fit$value - reduced_fit$value)
LR_p_value <- pchisq(LR_statistic, df = 1, lower.tail = FALSE)
c(statistic = LR_statistic, df = 1, p_value = LR_p_value)
```

```
   statistic           df      p_value
5.590288e+01 1.000000e+00 7.614106e-14
```

The likelihood-ratio statistic is approximately 55.90, with a very small p-value. There is very strong evidence of an association between sex and survival among third-class passengers after accounting for age group.

(d) Question from the maritime research manager

For a third-class adult female, $\mathbf{x}_0 = (1, 1, 1)^\top$. We first form the interval for $\eta_0 = \mathbf{x}_0^\top \boldsymbol{\beta}$, then transform its endpoints with Φ.

```
x0 <- c(1, 1, 1)
eta0_hat <- drop(x0 %*% beta_hat)
eta0_se <- sqrt(drop(t(x0) %*% variance_beta %*% x0))
eta0_interval <- eta0_hat + c(-1, 1) * z975 * eta0_se

c(
  probability = pnorm(eta0_hat),
  lower_95 = pnorm(eta0_interval[1]),
  upper_95 = pnorm(eta0_interval[2])
)
```

```
probability    lower_95    upper_95
  0.4460729   0.3750335   0.5188916
```

The model estimates a survival probability of about 0.446, with a 95% confidence interval of approximately 0.375 to 0.519. This interval measures uncertainty in the model's estimated mean probability for this group; it does not explain why an individual passenger survived.

Question 2: Logistic regression with Antarctic penguins

Data preparation

```
library(palmerpenguins)
data(penguins)

penguin_analysis <- na.omit(
  penguins[c("sex", "body_mass_g")]
)
penguin_analysis$male <-
  as.integer(penguin_analysis$sex == "male")
str(penguin_analysis)
```

```
tibble [333 × 3] (S3: tbl_df/tbl/data.frame)
 $ sex        : Factor w/ 2 levels "female","male": 2 1 1 1 2 1 2 1 2 2 ...
 $ body_mass_g: int [1:333] 3750 3800 3250 3450 3650 3625 4675 3200 3800 4400 ...
 $ male       : int [1:333] 1 0 0 0 1 0 1 0 1 1 ...
 - attr(*, "na.action")= 'omit' Named int [1:11] 4 9 10 11 12 48 179 219 257 269 ...
  ..- attr(*, "names")= chr [1:11] "4" "9" "10" "11" ...
```

(a) Change in log-odds

For the simple model,

$$\log\left\{\frac{p(x)}{1 - p(x)}\right\} = \beta_0 + \beta_1 x.$$

Therefore,

$$\log\{\text{odds}(x + 500)\} - \log\{\text{odds}(x)\} = 500\beta_1.$$

```
simple_penguin <- glm(
  male ~ body_mass_g,
  family = binomial(link = "logit"),
  data = penguin_analysis
)
summary(simple_penguin)
```

```

Call:
glm(formula = male ~ body_mass_g, family = binomial(link = "logit"),
    data = penguin_analysis)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -5.1625416  0.7243906  -7.127 1.03e-12 ***
body_mass_g  0.0012398  0.0001727   7.177 7.10e-13 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 461.61  on 332  degrees of freedom
Residual deviance: 396.64  on 331  degrees of freedom
AIC: 400.64

Number of Fisher Scoring iterations: 4
```

```
mass_beta_simple <- coef(simple_penguin)["body_mass_g"]
500 * mass_beta_simple
```

```
body_mass_g
  0.6199096
```

The estimated change in log-odds for a 500 g increase is 0.620.

(b) Confidence interval and interpretation

```
mass_se_simple <- sqrt(
  vcov(simple_penguin)["body_mass_g", "body_mass_g"]
)

log_odds_500 <- 500 * c(
  estimate = mass_beta_simple,
  lower_95 = mass_beta_simple - z975 * mass_se_simple,
  upper_95 = mass_beta_simple + z975 * mass_se_simple
)
log_odds_500
```

```
estimate.body_mass_g lower_95.body_mass_g upper_95.body_mass_g
           0.6199096            0.4506296            0.7891896
```

The 95% CI of the estimated change in log-odds for a 500 g increase is 0.451 to 0.789.

The exponentiated estimate and confidence interval from part (a) are:

```
exp(log_odds_500)
```

```
estimate.body_mass_g lower_95.body_mass_g upper_95.body_mass_g
            1.858760             1.569300             2.201612
```

The estimated odds of a male record are multiplied by 1.86 (95% CI 1.57 to 2.20) for each 500 g increase in body mass. Thus, the data provide evidence that heavier penguins have higher odds of being recorded as male.

This simple model does not account for species, even though penguin species differ in typical body mass and may also differ in the relationship between body mass and recorded sex. The result is an association and should not be interpreted causally or used as a stand-alone field classification rule.

Question 3: Gamma theory and New York ozone modelling

(a) Exponential-family form

The gamma log-density is

$$\log f(y; \nu, \lambda) = (\nu - 1) \log y - \lambda y + \nu \log \lambda - \log \Gamma(\nu).$$

Set

$$\phi = \frac{1}{\nu}, \qquad \theta = -\frac{\lambda}{\nu} = -\frac{1}{\mu}.$$

Then $a(\phi) = \phi, b(\theta) = -\log(-\theta)$, and one valid choice is

$$c(y, \phi) = \left(\frac{1}{\phi} - 1\right) \log y - \frac{\log \phi}{\phi} - \log \Gamma\left(\frac{1}{\phi}\right).$$

Substitution gives

$$f(y; \theta, \phi) = \exp\left[\frac{y\theta - b(\theta)}{\phi} + c(y, \phi)\right].$$

The parameter space is $y > 0, \theta < 0$, and $\phi > 0$.

(b) Canonical link and variance function

For an exponential dispersion family,

$$E(Y) = b'(\theta), \qquad \text{Var}(Y) = a(\phi) v(\mu).$$

Here

$$b'(\theta) = -\frac{1}{\theta} = \mu, \qquad b''(\theta) = \frac{1}{\theta^2}.$$

So the canonical link is

$$g(\mu) = \theta = -1/\mu,$$

and the variance function is

$$v(\mu) = b''((b')^{-1}(\mu)) = \mu^2.$$

Therefore,

$$E(Y) = \mu = \frac{\nu}{\lambda}, \qquad \text{Var}(Y) = \phi\mu^2 = \frac{\nu}{\lambda^2}.$$

Software often offers the positive inverse link $1/\mu$ because it is more convenient to express, although it is the negative of the canonical parameter.